OOPs Shape Calculator

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OOPs Mini Project (PPF)

Introduction

A calculator for the area and perimeter of Circle, Rectangle and Triangle through a common interface.

Class Design

Shape is the abstract base class with pure virtual area(), perimeter() and getName(). Circle, Rectangle and Triangle are derived classes, each with private data members and a constructor.

OOPs concepts used

1. **Classes and objects:** each shape is a class, and new Circle(r) creates an object.
2. **Encapsulation:** dimensions are private and set only through constructors.
3. **Constructors:** each shape initialises its own dimensions.
4. **Inheritance:** Circle, Rectangle and Triangle inherit from Shape.
5. **Virtual functions and overriding:** each derived class overrides area() and perimeter().
6. **Runtime polymorphism:** Shape *arr[3] holds all shapes, and arr[i]→display() calls the correct version at runtime.

Formulas

- **Circle→** area = πr^2
  perimeter = $2\pi r$
- **Rectangle→** area = $l \times w$
  perimeter = $2(l+w)$
- **Triangle→** area = $\sqrt{s(s-a)(s-b)(s-c)}$ (Heron's formula)
  perimeter = $a+b+c$.

Input validation

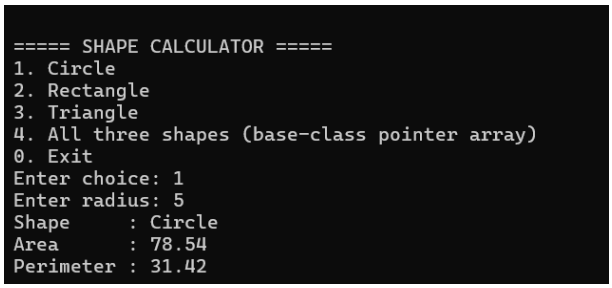
- Values must be greater than 0.
- Non-numeric input is rejected.
- A triangle must satisfy the sum of any two sides > the third.
- Menu choices outside 0 to 4 are rejected.

Test cases

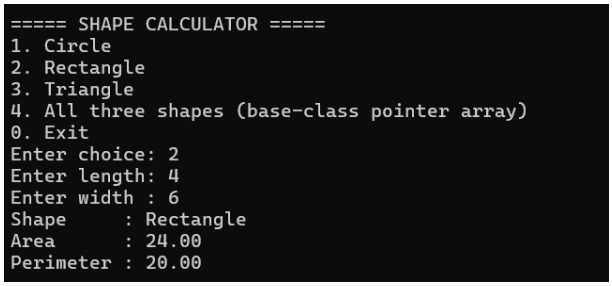
No.	Test	Input	Expected Result
1	Circle	Radius 5	Area 78.54, Perimeter 31.42
2	Rectangle	Length 4, Width 6	Area 24.00, Perimeter 20.00
3	Triangle	Sides 3, 4, 5	Area 6.00, Perimeter 12.00
4	Invalid input	-5, abc, 0, then 2	“Invalid input” three times, then Area 12.57, Perimeter 12.57
5	Invalid triangle	Sides 1, 2, 10, then 3, 4, 5	“Invalid triangle” message, then Area 6.00, Perimeter 12.00
6	Polymorphism	Option 4: radius 1, rectangle 2×3, triangle 3-4-5	Circle 3.14 / 6.28, Rectangle 6.00 / 10.00, Triangle 6.00 / 12.00
7	Invalid menu choice	7	“Invalid choice. Enter 0 to 4.”

Test Case Screenshots

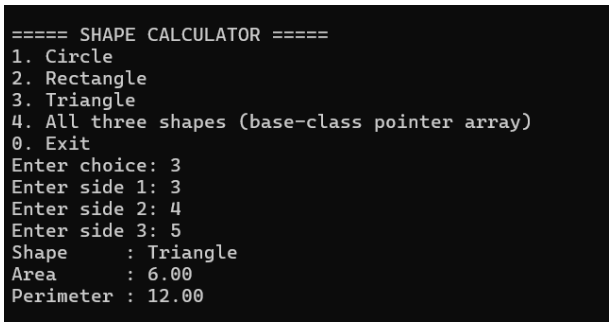
1. Test 1: Circle



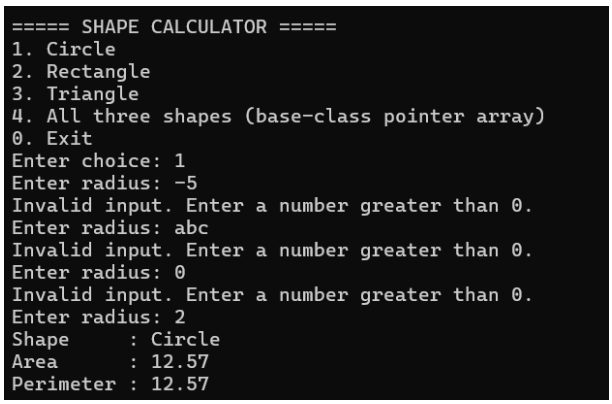
2. Test 2: Rectangle



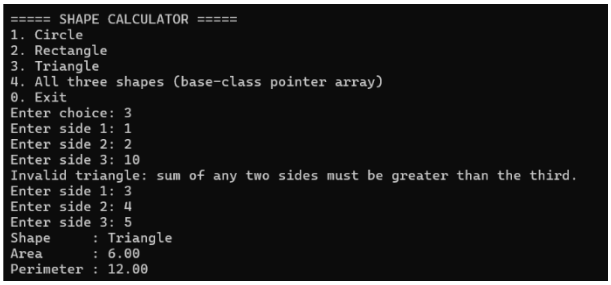
3. Test 3: Triangle



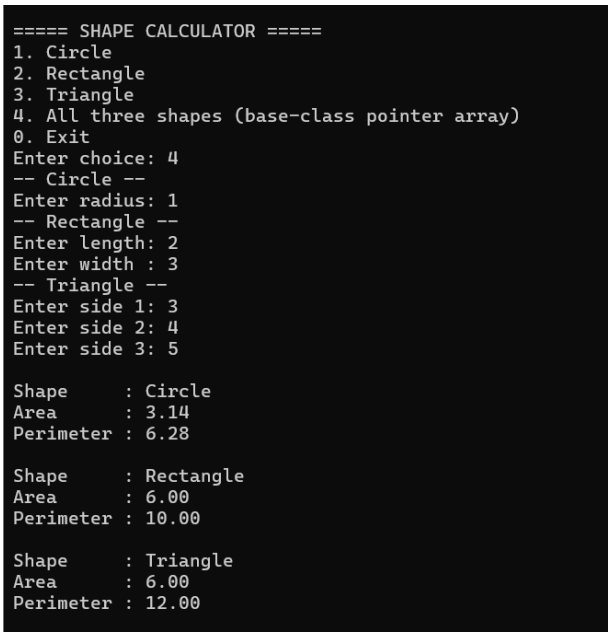
4. Test 4: Invalid input



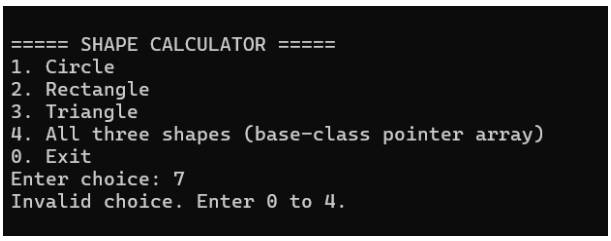
5. Test 5: Invalid triangle



6. Test 6: Runtime polymorphism



7. Test 7: Invalid menu choice



Conclusion

The project calculates the area and perimeter of a circle, rectangle and triangle using an abstract base class Shape and three derived classes. It demonstrates inheritance, encapsulation, constructors, virtual functions, function overriding and runtime polymorphism through a base-class pointer. Inputs are validated, and the program was tested with normal, boundary and invalid values.